

Lesson 5 Practice

Period _____

2. In exponential form, the expression $\sqrt{x}$ is equivalent to $x^{\frac{1}{2}}$.

For questions 3-5, rewrite each exponential expression as a radical expression. Then, evaluate the expression.

3. $(-729)^{\frac{1}{3}} = \sqrt[3]{-729} = -9$

-729
 $\swarrow \searrow$
 $-9 \cdot -81$
 $\swarrow \searrow$
 $-9 \cdot -9 = -9$

4. $(243)^{\frac{1}{5}} = \sqrt[5]{243}$
 $= 3$

243

9 · 81

3 3

9 · 9

3 · 3 3 · 3

$= \sqrt[5]{3 \cdot 3 \cdot 3 \cdot 3 \cdot 3} = 3$

5. $(225)^{\frac{1}{2}} = \sqrt{225} = 15$

Factor tree for 225:

```
graph TD
    225 --> 5
    225 --> 45
    45 --> 5
    45 --> 9
    9 --> 3
    9 --> 3
```

$= \sqrt{3 \cdot 3 \cdot 5 \cdot 5}$
 $3 \cdot 5 = 15$

6. Simplify the expression using the laws of exponents and then find the value.

$$\begin{aligned} (-27)^{-\frac{2}{9}} \cdot (-27)^{\frac{5}{9}} &= (-27)^{-\frac{2}{9} + \frac{5}{9}} \\ &= (-27)^{\frac{3}{9}} \\ &= (-27)^{\frac{3 \div 3}{9 \div 3}} \\ &= (-27)^{\frac{1}{3}} \\ &= \sqrt[3]{-27} \\ &= \sqrt[3]{-3 \cdot -3 \cdot -3} \\ &= -3 \end{aligned}$$

The expression $\frac{(-125)^{\frac{11}{9}}}{(-125)^{\frac{8}{9}}}$ can be rewritten using the law of exponents.

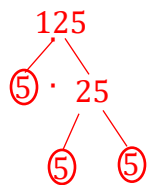
7. Which law of exponents can be used to rewrite the expression?
Quotient of Powers law

8. Rewrite the expression as an equivalent expression.

$$\begin{aligned} & (-125)^{\frac{11}{9} - \frac{8}{9}} \\ & (-125)^{\frac{3}{9}} \\ & = (-125)^{\frac{1}{3}} \end{aligned}$$

9. Write the simplified expression in radical form.
 $\sqrt[3]{-125}$

10. Evaluate the radical expression.



$$\begin{aligned} & -5 \cdot -5 \cdot -5 = 125 \\ & \sqrt[3]{-125} \\ & = \sqrt[3]{-5 \cdot -5 \cdot -5} \\ & \quad \text{= -5} \end{aligned}$$